

Improving Public Transport Modeling in MATSim: Fare Pricing and Routing Parameter Optimization

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Introduction

Public transport (PT) fares and route-choice preferences are two central determinants of PT mode choice, and therefore of model credibility for policy analysis such as fare reforms or service changes. MATSim’s scoring framework represents monetary cost only generically, via a per-distance rate and mode-specific constants, with no built-in notion of an operator’s actual pricing rules (Rieser, 2016). Route choice is handled by SwissRailRaptor’s generic transfer and waiting penalties. Existing MATSim extensions mostly target the latter, e.g. transfer penalties scaling with travel time or crowding-dependent in-vehicle disutility (Subbaraman, 2025).

As part of the redevelopment of the Swiss national agent-based transport model¹ (Dib et al., 2026), built with the eqasim pipeline (Hörl and Balac, 2021), we present two contributions: a data-driven calibration of SwissRailRaptor routing parameters against observed travel behavior, building on a CMA-ES implementation proposed by Sebastian Hörl², and a tariff-authority-aware PT fare model replacing eqasim’s default distance-based approximation. The fare model is currently specific to the Swiss tariff system, but its underlying methodology is transferable to any MATSim-based model.

Calibration of SwissRailRaptor routing parameters

We calibrate SwissRailRaptor’s mode-specific in-vehicle values of time and transfer penalties against observed PT travel behavior from the Swiss Microcensus on Mobility and Transport (Federal Office for Spatial Development and Federal Statistical Office, 2017), characterized by the number of legs per mode category (train; tram/metro; bus and other) and transfers per transfer type (train $\leftrightarrow$ train, train $\leftrightarrow$ other, other $\leftrightarrow$ other). As we are specifically interested in intermodality, we distinguish transfer penalties by transfer type, unlike the reference CMA-ES implementation, which treats all transfers identically. Initial values come from the mode choice model of Weis et al. (2017), a Swiss stated-preference study that did not distinguish PT sub-modes or intermodal transfers, so all three in-vehicle values of time and all three transfer penalties start from identical values.

Calibration minimizes the discrepancy between simulated and observed leg/transfer distributions: at each iteration, Microcensus trips are re-routed with the current parameters, and updates are proposed by the Covariance Matrix Adaptation Evolution Strategy (CMA-ES), a derivative-free method suited to this non-convex, noisy setting. Walk and wait values of time are excluded from calibration, as walk time is imprecisely recorded in the Microcensus and MATSim’s plan optimization leaves agents essentially no waiting time.

¹<https://github.com/eqasim-org/eqasim-switzerland/tree/cmdp>

²<https://github.com/eqasim-org/eqasim-switzerland/tree/cmdp>

²<https://github.com/eqasim-org/choice-model>

Table 1 reports the calibrated values. Relative to train, the in-vehicle value of time for bus and other modes increases substantially in magnitude, indicating lower perceived comfort, while tram/metro become marginally more attractive, consistent with their higher service quality. Transfer penalties show a clear hierarchy: train-to-train remains least penalized, intermodal transfers are penalized almost twice as much, and other-to-other transfers fall in between.

	Before calibration	After calibration
In-vehicle values of time [h^{-1}]		
train (reference)	-1.00	-1.00
tram + subway	-1.00	-0.95
bus + other	-1.00	-1.29
Other values of time [h^{-1}]		
walk (not calibrated)	-1.35	
wait (not calibrated)	-1.29	
Transfer penalties [-]		
train $\leftrightarrow$ train	0.09	0.07
train $\leftrightarrow$ other	0.09	0.16
other $\leftrightarrow$ other	0.09	0.12

Table 1: Public transport routing parameters, before and after optimization.

A tariff-authority-aware public transport fare model

Swiss public transport is operated by a large number of companies, organized in three broad tiers: SBB, the national railway, which operates most regional and long-distance rail services; regional fare associations, in which several local operators within a metropolitan region or canton jointly issue tickets under a common, area-based pricing system; and a constellation of smaller, independent operators, such as local bus companies and mountain railways, each generally setting its own fares.

We develop a three-component fare model mirroring this tiered organization, implemented in `eqasim-switzerland` (data preparation) and `eqasim-java` (fare computation at runtime). *(i)* SBB pricing is distance-based, but relies on standardized tariff distances, reconstructed from official reference documents into a weighted graph, that do not always exactly reflect real network distances due to surcharges on high-demand lines. *(ii)* The 18 regional fare associations instead price trips based on the number of fare zones crossed, whose structures and pricing rules were extracted into a flexible pricing schema (Figure 1). *(iii)* As the remaining independent operators are too numerous to model individually, their fares default to a single quadratic model of price against distance, estimated by web-scraping roughly 28700 fare observations ($R^2 = 0.855$).

At runtime, each PT trip’s leg sequence is traversed to identify the responsible pricing authority per segment via a priority-based heuristic, and the segment-level information is passed to a fare calculator that applies the appropriate model to return the trip’s total cost. This cost can be further adjusted by discounts customizable through the config, before being used directly in agents’ mode choice utility.

Conclusion and outlook

The two developments presented here address complementary gaps in MATSim’s default PT representation: a routing-parameter calibration grounding SwissRailRaptor’s route choice in observed Microcensus patterns, and a fare model reflecting Switzerland’s fragmented tariff landscape. While the fare model is Swiss-specific, its architecture and the routing calibration are

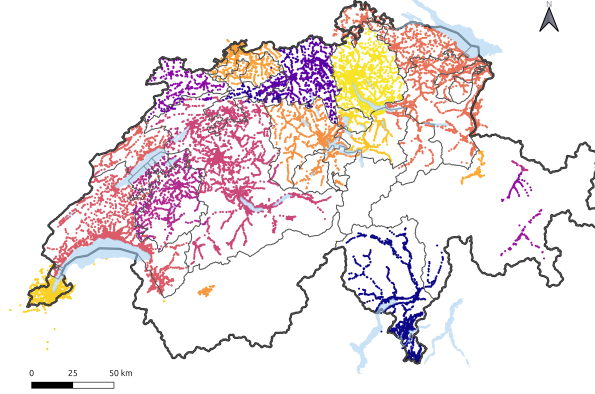


Figure 1: Reconstructed coverage areas of the 18 modeled regional fare associations.

directly transferable to other MATSim contexts. Future work on the fare model includes a better representation of subscriptions and day passes, currently not accounted for, and a more flexible design allowing pricing policies, such as age-based discounts, to be toggled easily for scenario analysis.

References

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